



REDROB · DATA & AI CHALLENGE

# HireFit Ranker

Ranks **careers**, not **keywords**.

A deterministic, offline, CPU-only engine that reads who actually fits a role — then returns a 100-candidate shortlist a recruiter can trust.

**100K**

candidates scored

**80s**

cloud CPU runtime ( $\leq 300$ s)

**0.894**

NDCG@10 \*

**0**

honeypots in top-100

\* NDCG@10 on an independent LLM-judged sample (dev proxy) — see slide 9. Not the hidden challenge score.

HIREFIT RANKER

## THE PROBLEM

# Keyword filters miss the right people — and rank the wrong ones first.

The dataset is engineered to punish keyword matching. Four trap types sink a naive ranker.



### Keyword stuffers

Wrong-role profiles padded with AI skills — a Marketing Manager listing 9 AI keywords.



### Wrong-role profiles

Perfect skill lists attached to titles that never shipped a ranking system.



### Honeypots

Subtly impossible profiles — “expert” in 10 skills with 0 months used. Forced to tier 0.



### Unavailable-but-perfect

Great on paper, but 5% recruiter response and not logged in for months.



**The real task:** rank candidates the way a great recruiter would — reading career history and behavior, not counting keywords. A candidate who *built* a recommender at a product company outranks one who merely *lists* “RAG” and “Pinecone.”

# The JD lists keywords; the role is hiring for evidence.

Our whole design follows from reading the gap between what the JD says and what it means.



## What the JD says

*“5–9 years, embeddings, retrieval, ranking, NLP, deep learning, PyTorch, TensorFlow...”*

embeddings

retrieval

ranking

NLP

5–9 yrs

**The trap:** skill-keyword overlap is high for stuffers and low for the engineers who actually shipped the systems.



## What the role means



**Shipped work** — Built & deployed ranking / search / recsys in production.



**Career trajectory** — Consistent growth in relevant, product-company roles.



**Right seniority** — anchored on the JD's 5–9 band — juniors demoted hard; senior depth read from trajectory, not years arithmetic.



**Actually available** — Responsive, open to work, reachable notice period.

# We measured the config that scores higher — and declined it on principle.

Every knob swept under a decision rule committed before the first run; the JD is the spec, not the judge.

0.896

SHIPPED (LLM-JUDGE)

0.926

FLOOR-OFF — BIASED  
SAMPLE, DECLINED

0

RANK CHANGES 0.25–0.55

11 / 11

PLANTED TRAPS CAUGHT



## How the rule decided

**Rule first** committed before any sweep run — ties break toward the JD's down-weight instruction

**Coverage guard** floor 0.85–1.00 configs excluded: <95% label coverage, selection-biased

**Winner** shipped config — identical top-100 across floors 0.25–0.55



## We built our strongest rival — it lost

A LambdaMART challenger built on our own 33 features plus recovered generator structure lost its pre-registered gate: 0.900 vs 0.906. One of four measured negatives — embeddings, learned-LR, the challenger, and a declined availability hedge. The adopted change is role-family depth scoring; an 8-swap calibration pass was trialed then removed — the official path now uses NO candidate-ID tuning (CI-enforced).



**Honest framing:** the pool plants 11 perfect-on-paper-but-unavailable gold profiles — one a honeypot in gold clothing — the guardrails demote all 11; an LLM judge missed one of them.

## THE SYSTEM

# Deterministic at rank time — every choice gated by LLM judges offline.

Six CPU-only stages, no network or GPU; tuned and frozen under two LLM judges and a LambdaMART challenger gate.



**Final score** = base score × behavioral × honeypot × disqualifier

Guardrails are multiplicative — a strong-on-paper profile that is impossible, off-target, or unreachable is pushed down no matter how good its keywords look.

RUNS UNDER:

✓ Offline

✓ CPU-only

✓ No GPU

✓  
Deterministic

✓ bm25s  
backend

# 33 features separate keyword lists from recruiter-plausible fit.

Each signal is computed deterministically from profile facts — and every rank decomposes back into them.



## Skills & depth

10 features

core / nice-to-have match, skill depth, endorsement trust, assessment scores, GitHub, semantic (HyRE) match



## Career evidence

11 features

product-company ratio, IR/ranking experience, production evidence, seniority, trajectory, scale, role-family depth (backend, data/BI)



## Experience fit

4 features

years-of-experience band, education, ML/AI tenure, open-source signal



## Behavior

6 features

availability, engagement, responsiveness, interview reliability, profile quality, notice



## Logistics

2 features

preferred-location match, willingness to relocate



## Multiplicative guardrails

Behavioral, honeypot & disqualifier multipliers gate the base score so traps can't buy their way to the top.

# The shortlist is real: every rank is traceable.

Live output from the released 100K pool, plus the exact feature values and guardrails behind the #1 candidate.

Ranked output

submission.csv top 6

| Rank | Candidate evidence                                                                     | YOE | Score |
|------|----------------------------------------------------------------------------------------|-----|-------|
| 1    | <b>Senior ML Engineer - Zomato</b><br>hybrid retrieval + ranking at 50M+ queries       | 7.2 | 1.000 |
| 2    | <b>Senior NLP Engineer - Niramai</b><br>semantic search + embeddings + strong response | 7.8 | 0.940 |
| 3    | <b>Senior ML Engineer - Genpact AI</b><br>IR, pgvector, Sentence Transformers          | 6.1 | 0.933 |
| 4    | <b>Senior AI Engineer - Apple</b><br>FAISS + fine-tuning + 30-day notice               | 5.9 | 0.931 |
| 5    | <b>Senior ML Engineer - Flipkart</b><br>OpenSearch + LLM fine-tuning; logistics flag   | 8.1 | 0.921 |
| 6    | <b>Senior AI Engineer - Netflix</b><br>Elasticsearch + Qdrant + PEFT                   | 7.8 | 0.902 |

**Readout:** the top six are senior AI/ML/NLP/search profiles with production evidence; tie-breaks are deterministic by candidate\_id.

Why #1 wins

CAND\_0018499

**Senior ML Engineer** at Zomato | 7.2 yrs | Noida | 15-day notice

core\_skill\_match

0.982

production\_evidence

1.000

ir\_ranking\_experience

1.000

product\_company\_ratio

1.000

career\_trajectory\_score

1.000

behavior x1.10

honeypot x1.00

DQ x1.00

**Grounded reason:** shipped retrieval/ranking; skills include Learning to Rank + Embeddings; response 0.61.

# Stuffers and honeypots never reach the shortlist.

Measured on the released pool — the guardrails do real work, not decoration.



Keyword stuffers buried

4,800+

Wrong-role profiles across the pool's 12 non-target strata — zero reach the top-100.



Honeypots blocked

53 → 0

Impossible profiles detected, then zero emitted in the top 100. Challenge DQ threshold is >10%.



Unavailable down-weighted

×0.42

Behavioral multiplier on an 8-yr profile with 12% response & not-open-to-work — pushed off the shortlist.



**Plain-language fit is captured too:** skill matching runs over the full profile text, not just the skills array — so an engineer who *describes* building a recommender (without the buzzwords) still scores on core fit.



## VALIDATION

# Three independent LLM judge families confirm the top-10 — it isn't self-graded.

We avoided grading our ranker with its own heuristics. A separate judge scored a stratified sample.

0.894

NDCG@10

0.873

NDCG@50

0.912

MAP

1.00

P@10



### How it was judged

**249** stratified candidates — ranker top ranks + higher heuristic tiers + a random control

**Judges** gemini-2.5-flash + gpt-4.1-mini + deepseek, same rubric, same 249 ids: kappa 0.918–0.935 across all pairs.

**Top-10** all 4–5 (judge 1); all tier-5, NDCG@10 = 1.0 (judge 2)



### Not propped up by our own labels

The judge composite (0.896) is slightly higher than our non-circular heuristic harness (0.886) — the ranking is strong by an outside opinion, not just our own. Two independent cross-family judges (gpt-4.1, gemini-2.5-pro) — never used to select it — also confirm the shipped hedge beats golden (+0.020 / +0.016).



**Honest framing:** a development-time proxy on a 249-candidate sample, not the hidden challenge score. An 8-swap calibration pass was trialed (judge re-test +0.0124) but later removed; the shipped ranking uses no candidate-ID tuning (docs/llm\_judge\_eval\_3.md).

# Reproducibility beats raw model power.

The deck does not hide limitations; it turns them into Stage-5 defense points judges can verify.

## Defensible decisions

shipped path

- **No LLM per candidate**  
100K hosted calls break the CPU/no-network budget and make reproduction fragile.
- **Dense embeddings tested, then kept off**  
NDCG@10 was flat while runtime and complexity rose, so the simpler ranker won.
- **Guardrails are multiplicative**  
Impossible, off-target, or unreachable profiles cannot buy rank with keywords.
- **Deterministic reasoning**  
Same input gives the same CSV; every sentence points to real profile facts.

## Next hardening moves

honest roadmap

- **Hidden ground truth**  
The official leaderboard is unseen, so proxy eval is evidence - not a score claim.
- **Conservative hard-trap recall**  
53 hard traps are blocked; softer suspicious profiles are down-weighted, not over-zeroed.
- **Learned ranking layer**  
Future labels could train LambdaMART/XGBoost over these same features.
- **Stage-3 fallback**  
If multiprocessing is constrained, --workers 1 preserves the exact same ranking output.

## REPRODUCIBILITY

# One command reproduces the shortlist — offline, well under the five-minute limit.

Built for the Stage-3 sandbox: it runs unmodified inside the challenge's compute box.

```
>_ python rank.py --candidates ./candidates.jsonl --out ./submission.csv
```

### CONSTRAINT

### CHALLENGE LIMIT

### MEASURED

**Runtime**

≤ 5 min (300s)

**80–125 s**



**Memory**

≤ 16 GB RAM

**~6.1 GB peak**



**Compute**

CPU only

**CPU only**



**Network**

off during ranking

**fully offline**



### Submission package



GitHub repo — clean, tested, reproducible



submission.csv — top-100 + grounded reasoning



METHODOLOGY.md — every choice, with the why



Live sandbox — HuggingFace Space



Independent eval evidence committed (249 labels)

# A drop-in scoring layer for Redrob's hiring stack — auditable by design.

Ranks talent on behavior and evidence, not resumes — the same thesis as the platform it would join.

**~3 ms**

AUDIT API, MEASURED

**1 cmd**

FULL REPRODUCTION

**0**

LLM CALLS PER RANKED  
CANDIDATE

**33**

EXPLAINABLE FEATURES



## Integration path

**JD compiler** any role text compiles into the same deterministic scoring program

**Audit payload** per-candidate JSON: features, multipliers, flags — recruiter-readable

**Signals** behavioral multipliers consume Redrob activity data natively



## Roadmap, honestly

Indic-script normalization is a localized swap (one module) — the architecture is script-agnostic. When real outcome labels exist, LambdaMART can replace the hand weights over the same 28 features without touching the audit trail: the features are the durable asset.



**LLMs where they are strong — offline judging and evaluation — and deterministic evidence where hiring decisions need to be defended.**

## THE TAKEAWAY

# We don't rank resumes. We rank **hireability**.



Reads careers & behavior, not buzzwords



Every rank explained by 33 named features



Top-10 verified by an independent judge



Blocks honeypots, stuffers & the unavailable



100K scored offline, on CPU, in 80s on cloud hardware